Arjun Sharma

Senior Machine Learning Engineer

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Summary

Senior ML Engineer with 6+ years of experience building and deploying large-scale machine learning systems at top tech companies. Deep expertise in recommendation systems, NLP, and MLOps. Proven track record of improving model performance and driving measurable business impact across ads, search, and personalization products.

Experience

Senior Machine Learning Engineer — Google (Search & Recommendations)

Mountain View, CA | Jun 2021 – Present

- **Recommendation Systems:** Designed and deployed a two-tower neural retrieval model for YouTube video recommendations, improving click-through rate by **18%** and watch time by **12%** across 2B+ daily active users
- **Model Lifecycle Management:** Built an end-to-end MLOps pipeline using TFX, Vertex AI, and Kubeflow that reduced model deployment time from 2 weeks to **under 4 hours**, enabling 3× faster experimentation cycles
- **LLM Fine-tuning:** Led fine-tuning of a 7B-parameter language model on internal search queries using RLHF; reduced irrelevant search results by **22%** as measured by NDCG@10
- **A/B Testing Infrastructure:** Designed a distributed feature store (using BigTable + Feast) that served real-time features to 50+ ML models with **<5ms P99 latency**
- **Team Leadership:** Mentored 4 junior ML engineers; ran weekly ML paper reading sessions adopted by the broader Search org (~80 people)

Machine Learning Engineer II — Meta (Ads & Monetization)

Menlo Park, CA | Aug 2019 – May 2021

- **Ads Ranking:** Retrained and optimized Meta's click prediction model (DLRM architecture) using sparse embedding compression, reducing memory footprint by **35%** without degrading AUC

- **Feature Engineering:** Automated feature importance analysis using SHAP values at scale, cutting irrelevant feature count by 40% and speeding up model training by 1.3×
- **Calibration & Fairness:** Implemented isotonic regression-based probability calibration for ads CTR prediction; improved log-loss from 0.42 → 0.39 across multiple audience segments
- **Cross-functional Collaboration:** Partnered with product, data engineering, and infra teams to ship a real-time bidding model improvement that generated **$120M+ incremental annualized revenue**

Data Scientist / ML Engineer — Swiggy (Bengaluru, India)

Jul 2017 – Jul 2019

- Built a demand forecasting model (XGBoost + LSTM ensemble) for food delivery ETA prediction; reduced average ETA error by 28%, improving customer satisfaction scores by 15%
- Developed an RFM-based customer segmentation pipeline processing 5M+ orders/day; enabled targeted push notifications that improved 30-day retention by 9%
- Shipped an NLP-based restaurant review classifier (BERT fine-tuned) for content moderation, achieving **94.2% accuracy** on a 100K-sample test set

Education

M.S. in Computer Science (Machine Learning Specialization)
Stanford University, Stanford, CA | 2015 – 2017 | GPA: 3.92/4.0

B.Tech in Computer Science
IIT Bombay, Mumbai, India | 2011 – 2015 | GPA: 9.1/10.0

Skills & Technologies

Category	Technologies
ML Frameworks	PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace Transformers
MLOps & Infra	Kubeflow, MLflow, Vertex AI, TFX, Docker, Kubernetes, Apache Airflow
Distributed Computing	Spark, Ray, Dask, BigQuery, Dataflow
Cloud Platforms	GCP (primary), AWS (SageMaker, S3, EC2), Azure ML
Languages	Python (expert), C++ (proficient), SQL, Go (familiar)
Specializations	Recommendation Systems, NLP/LLMs, Ads Ranking, Computer Vision

Selected Projects & Publications

- **Paper (NeurIPS 2022):** "Efficient Sparse Embedding Compression for Large-Scale Recommendation Models" — 140+ citations
- **Open Source:** Contributed to HuggingFace `transformers` (PR #18431) — added mixed-precision support for T5 fine-tuning; 2.3k GitHub stars on personal `ml-feature-store` repo
- **Kaggle:** Kaggle Master (top 0.2%); 1st place finish in the Google Brain — Ventilator Pressure Prediction competition (2021)

Awards & Recognition

- Google **"Spot Bonus"** recipient (2022) — for impact on Search quality improvements
- Meta **"Above & Beyond" award** (2020) — for ads revenue impact project
- Graduate Research Fellowship, Stanford University (2015–2017)

Interests

- AI Safety & Alignment (active contributor to EleutherAI discussions)
- Badminton (state-level, India) | Long-distance running | Reading: history of science

References available upon request